

Initiating Search

November 4, 2025, 3:53 PM

• Search:

CAS Lexicon Search:

Nozaki-Hiyama-Kishi reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (231)	 References	View Results
Exported: Retrieved Related Reaction Results (1,082)	 Reactions	View Results

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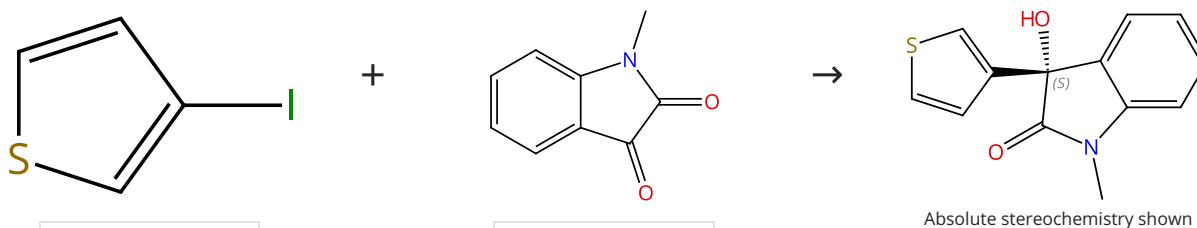
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (68)

 Suppliers (88)

Absolute stereochemistry shown

31-614-CAS-44352343

Steps: 1 Yield: 99%

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

Chinese Chemical Letters (2025), 36(5), 110243.

1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron

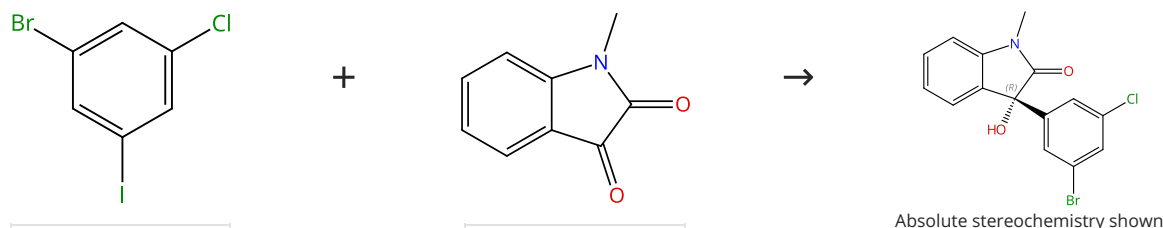
Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]

Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (75)

 Suppliers (88)

Absolute stereochemistry shown

31-614-CAS-44352325

Steps: 1 Yield: 99%

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

Chinese Chemical Letters (2025), 36(5), 110243.

1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron

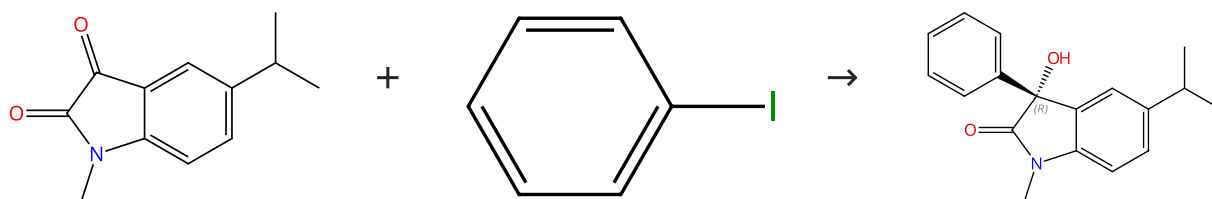
Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]

Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (2)

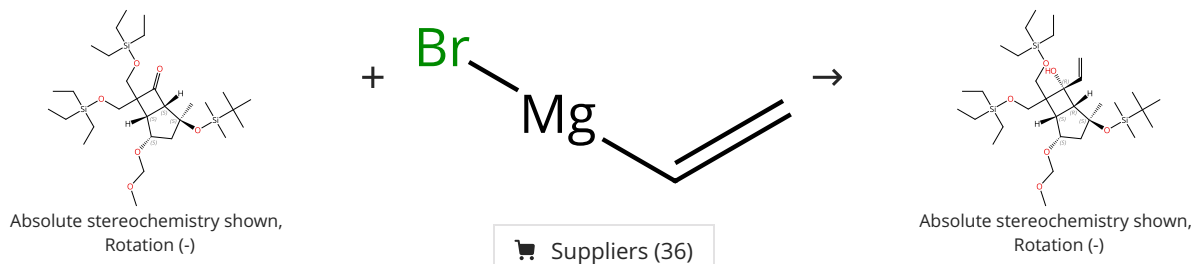
 Suppliers (95)

Absolute stereochemistry shown

31-614-CAS-44352280	Steps: 1 Yield: 99%	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis
1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1 <i>R</i> ,1' <i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 24 h, 85 °C		By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
Experimental Protocols		

Scheme 4 (1 Reaction)

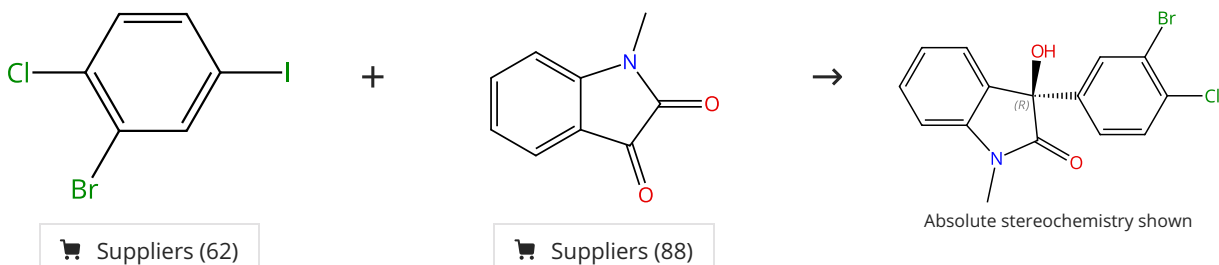
Steps: 1 Yield: 99%



31-320-CAS-4641246	Steps: 1 Yield: 99%	An Approach to the Carbon Backbone of Bielschowskysin, Part 2: Non-Photochemical Strategy
1.1 Solvents: Diethyl ether; overnight, rt		By: Farcet, Jean-Baptiste; et al European Journal of Organic Chemistry (2013), 2013(36), 8245-8252.
Experimental Protocols		

Scheme 5 (1 Reaction)

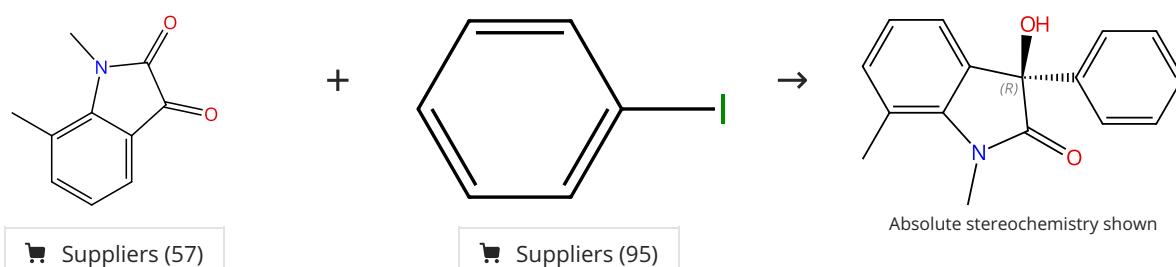
Steps: 1 Yield: 99%



31-614-CAS-44352320	Steps: 1 Yield: 99%	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis
1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1 <i>R</i> ,1' <i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 24 h, 85 °C		By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
Experimental Protocols		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 99%



31-614-CAS-44352281 Steps: 1 Yield: 99% 1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1<i>R</i>,1'<i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 24 h, 85 °C Experimental Protocols	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
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Scheme 7 (1 Reaction)

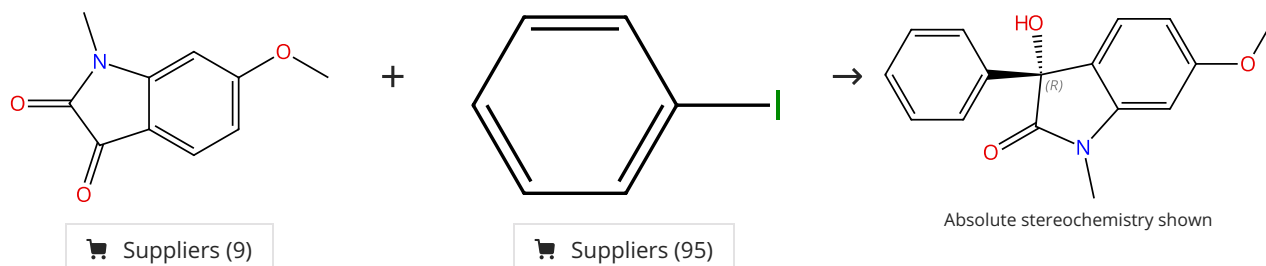
Steps: 1 Yield: 99%



31-320-CAS-18334469 Steps: 1 Yield: 99% 1.1 Reagents: Zinc Solvents: Diethyl ether, Dimethylformamide, Toluene; 10 min, rt; 12 h, rt	Scalable Regioselective and Stereoselective Synthesis of Functionalized (E)-4-Iodobut-3-en-1-ols: Gram-Scale Total Synthesis of Fungal Decanolides and Derivatives By: Sherwood, Alexander M.; et al Journal of Organic Chemistry (2018), 83(2), 980-992.
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Scheme 8 (1 Reaction)

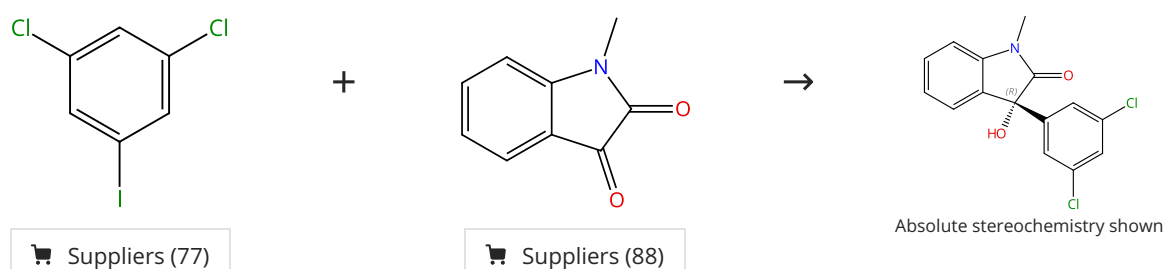
Steps: 1 Yield: 98%



31-614-CAS-44352286 Steps: 1 Yield: 98% 1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1<i>R</i>,1'<i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 48 h, 85 °C Experimental Protocols	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
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Scheme 9 (1 Reaction)

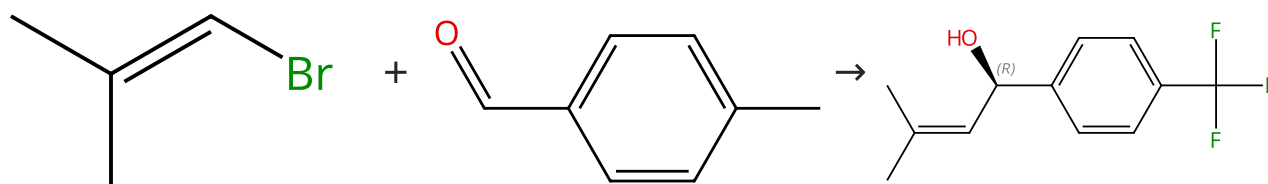
Steps: 1 Yield: 98%



31-614-CAS-44352316	Steps: 1 Yield: 98%	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis
1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1 <i>R</i> ,1' <i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 24 h, 85 °C		By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (41)

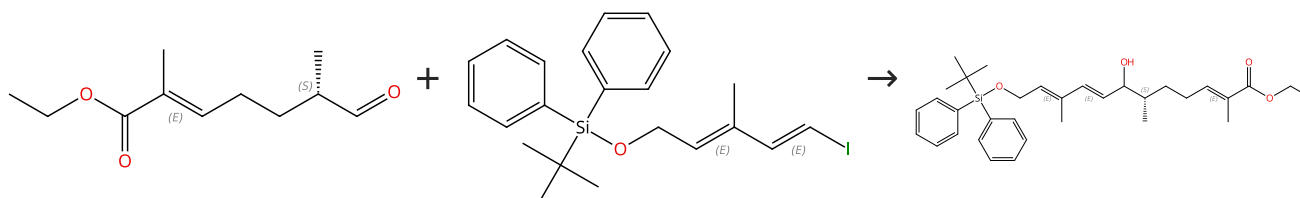
Suppliers (104)

Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-41431856	Steps: 1 Yield: 98%	Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis
1.1 Reagents: Hantzsch ester, Cyclohexylmorpholine, Zinc iodide Catalysts: Dibromo[1,2-di(methoxy-κO)ethane]nickel, 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile, 3054393-69-7 Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C		By: Gu, Pei; et al Angewandte Chemie, International Edition (2024), 63(37), e202408195.
Experimental Protocols		

Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%

Absolute stereochemistry shown
Double bond geometry shown

Double bond geometry shown

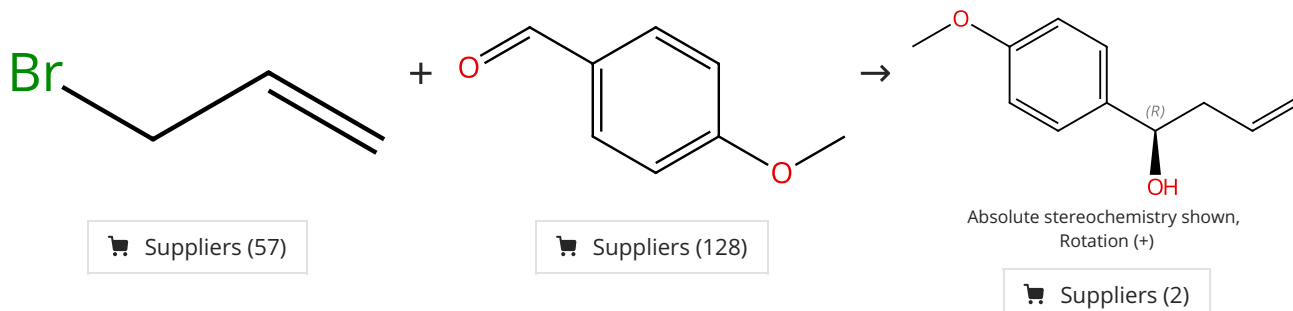
Absolute stereochemistry shown
Double bond geometry shown

Suppliers (2)

31-320-CAS-9551508	Steps: 1 Yield: 98%	Stereoselective synthesis of the southern domain of tubelactomicin A by a tandem intramolecular Diels-Alder/lactonization reaction
1.1 Reagents: Nickel dichloride Catalysts: Chromium dichloride Solvents: Dimethyl sulfoxide, Tetrahydrofuran; rt		By: Kim, Ji Yoon; et al Bulletin of the Korean Chemical Society (2009), 30(2), 289-290.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 98%



31-320-CAS-11561841

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Manganese
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 **Catalysts:** Triethylamine, 2,4-Bis(1,1-dimethylethyl)-6-[(*E*)-[[5*S*,7*R*,8*S*]-5,6,7,8-tetrahydro-6,6-dimethyl-2-phenyl-5,7-methanoquinolin-8-yl]imino]methyl]phenol; 1 h, rt
- 1.3 1 h, rt
- 1.4 **Reagents:** Chlorotrimethylsilane; 20 h, rt
- 1.5 **Reagents:** Sodium bicarbonate
Solvents: Water; rt
- 1.6 **Reagents:** Hydrochloric acid
Solvents: Tetrahydrofuran; rt; 30 min, rt
- 1.7 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

Experimental Protocols

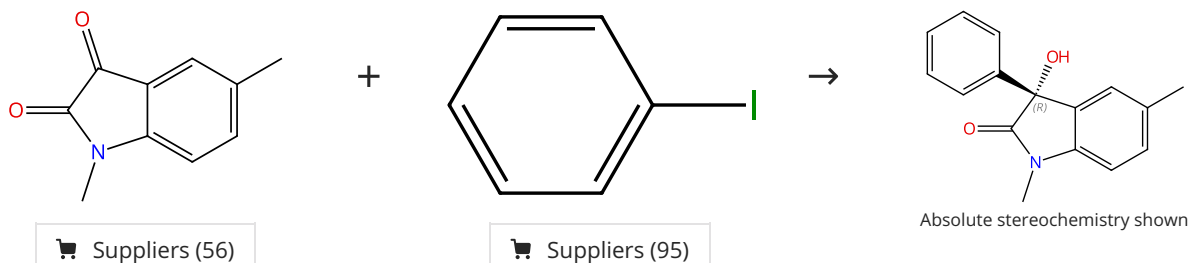
A pineno-salen type catalyst for the enantioselective Nozaki-Hiyama-Kishi reaction

By: Dhondge, Attrimuni P.; et al

Tetrahedron: Asymmetry (2012), 23(10), 723-733.

Scheme 13 (2 Reactions)

Steps: 1 Yield: 96-98%



31-614-CAS-44352277

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron
Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (*T*-4), (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]
Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

Chinese Chemical Letters (2025), 36(5), 110243.

31-614-CAS-44352351

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron
Catalysts: Cobalt, dichloro[(1*S*,1'*S*)-1,1'-(1,2-phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine-κP]]-
Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

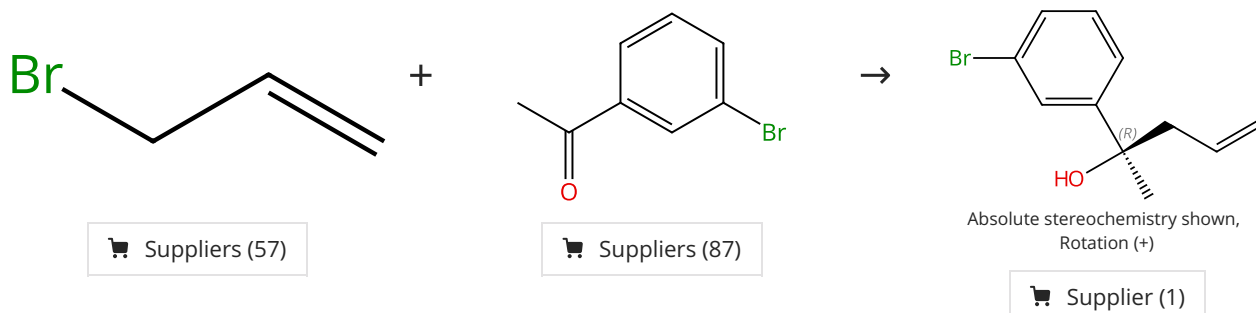
Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

Chinese Chemical Letters (2025), 36(5), 110243.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



31-320-CAS-5252402

Steps: 1 Yield: 98%

A Chiral Bipyridyl Alcohol for Catalytic Enantioselective Nozaki-Hiyama-Kishi Allylation of Aldehydes and Ketones

By: Chen, Rui-Yu; et al

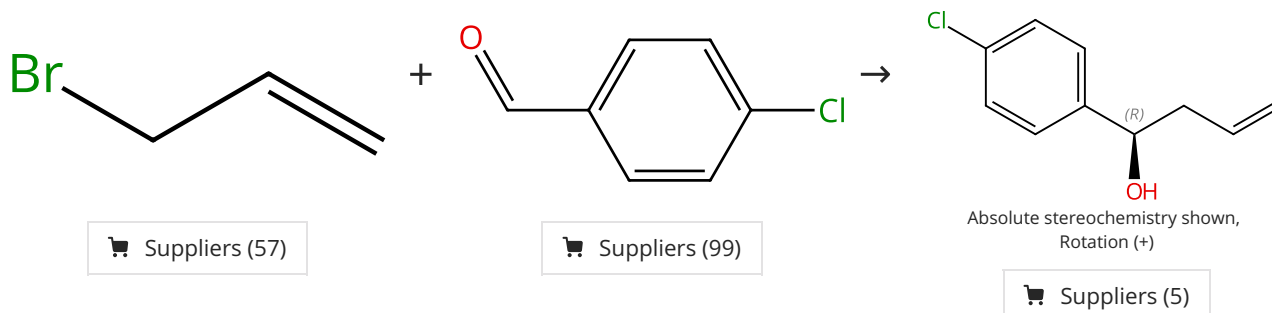
Advanced Synthesis & Catalysis (2015), 357(5), 961-966.

- 1.1 Reagents: Manganese
Catalysts: Chromium chloride (CrCl₃)
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 Reagents: Triethylamine
Catalysts: (5*S*,7*S*,8*R*)-5,6,7,8-Tetrahydro-6,6-dimethyl- α , α -diphenyl-2-(2-pyridinyl)-5,7-methanoquinoline-8-methanol; 1 h, rt
- 1.3 1 h, rt
- 1.4 Reagents: Chlorotrimethylsilane; 36 h, rt
- 1.5 Reagents: Sodium bicarbonate
Solvents: Water
- 1.6 Reagents: Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; rt; 15 min, rt
- 1.7 Reagents: Water

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



31-320-CAS-4078319

Steps: 1 Yield: 98%

Application of tridentate bis(oxazoline) ligands in catalytic asymmetric Nozaki-Hiyama allylation and crotylation: an example of high enantioselection with a non-symmetric bis(oxazoline) ligand

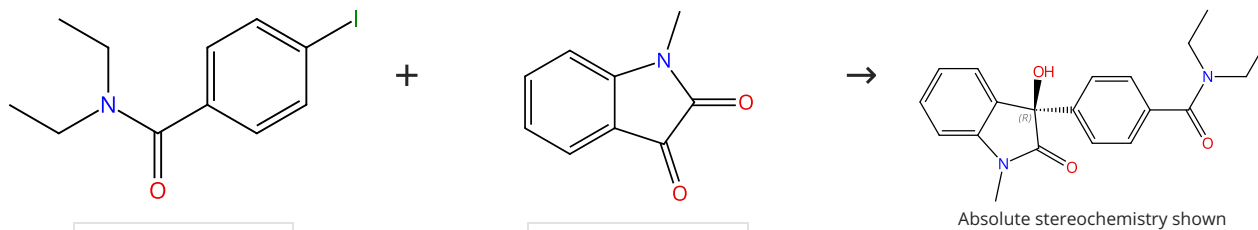
By: McManus, Helen A.; et al

Advanced Synthesis & Catalysis (2006), 348(4 + 5), 551-558.

- 1.1 Reagents: Manganese
Catalysts: Chromium chloride (CrCl₃)
Solvents: Acetonitrile, Tetrahydrofuran; 10 min; 60 min, rt
- 1.2 Reagents: Diisopropylethylamine
Catalysts: 2-[(4*S*)-4,5-Dihydro-4-(phenylmethyl)-2-oxazolyl]-*N*-[2-[(4*S*)-4-(1,1-dimethylethyl)-4,5-dihydro-2-oxazolyl]phenyl]benzenamine; 1 h, rt
- 1.3 1 h, rt
- 1.4 Reagents: Chlorotrimethylsilane; 16 h, rt
- 1.5 Reagents: Sodium bicarbonate
Solvents: Water
- 1.6 Reagents: Hydrochloric acid
Solvents: Tetrahydrofuran, Water; 5 min

Scheme 16 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (46)

Suppliers (88)

31-614-CAS-44352324

Steps: 1 Yield: 97%

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

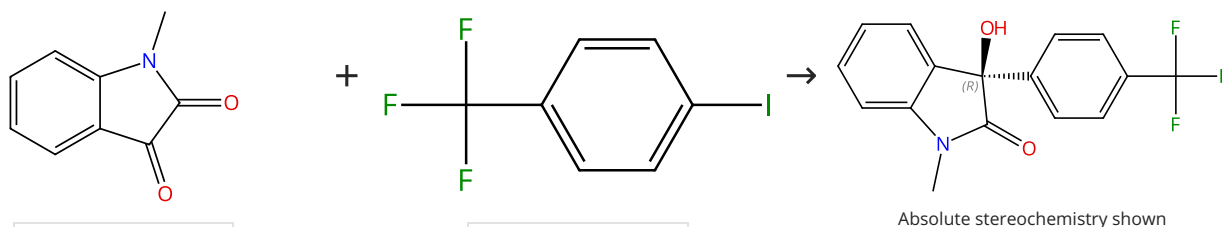
Chinese Chemical Letters (2025), 36(5), 110243.

1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron**Catalysts:** Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]**Solvents:** Mesitylene; 48 h, 85 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (88)

Suppliers (81)

31-614-CAS-44352327

Steps: 1 Yield: 97%

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

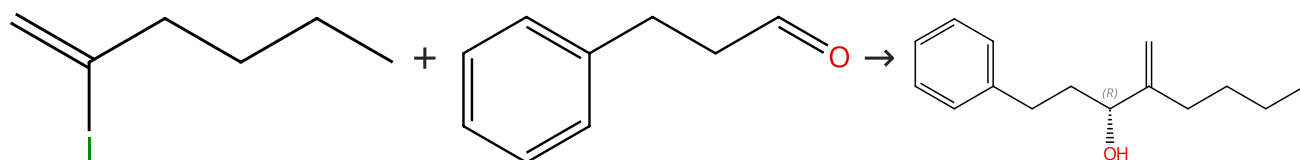
Chinese Chemical Letters (2025), 36(5), 110243.

1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron**Catalysts:** Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]**Solvents:** Mesitylene; 24 h, 85 °C

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 97%



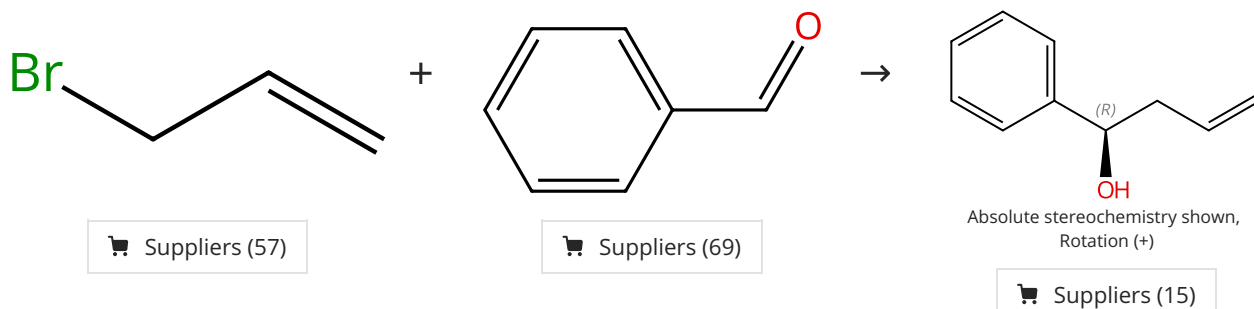
Suppliers (3)

Suppliers (90)

31-320-CAS-23456632	Steps: 1 Yield: 97%	Chromium(II) chloride By: Takai, Kazuhiko e-EROS Encyclopedia of Reagents for Organic Synthesis (2016), 1-19.
1.1	Reagents: Zirconocene, dichloride, Manganese, Lithium chloride, Proton sponge Catalysts: Chromium dichloride, (7-4)-Dichloro[3,5-dichloro- <i>N</i> -[2-[(4 <i>S</i>)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]-6-[4-(9-methyl-1,10-phenanthroline-2-yl-κ ^N ¹ ,κ ^N ¹⁰)butoxy]phenyl] benzenesulfonamide]nickel Solvents: Acetonitrile; 36 h, rt	

Scheme 19 (2 Reactions)

Steps: 1 Yield: 97%

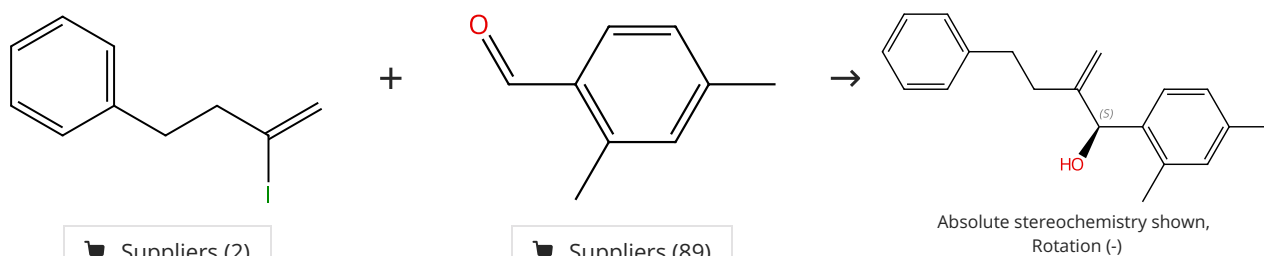


31-320-CAS-619306	Steps: 1 Yield: 97%	A pineno-salen type catalyst for the enantioselective Nozaki-Hiyama-Kishi reaction By: Dhondge, Attrimuni P.; et al Tetrahedron: Asymmetry (2012), 23(10), 723-733.
1.1	Reagents: Manganese Solvents: Tetrahydrofuran; 1 h, rt	
1.2	Catalysts: Triethylamine, 2,4-Bis(1,1-dimethylethyl)-6-[(<i>E</i>)-[[5 <i>S</i> ,7 <i>R</i> ,8 <i>S</i>]-5,6,7,8-tetrahydro-6,6-dimethyl-2-phenyl-5,7-methanoquinolin-8-yl]imino]methyl]phenol; 1 h, rt	
1.3	1 h, rt	
1.4	Reagents: Chlorotrimethylsilane; 20 h, rt	
1.5	Reagents: Sodium bicarbonate Solvents: Water; rt	
1.6	Reagents: Hydrochloric acid Solvents: Tetrahydrofuran; rt; 30 min, rt	
1.7	Reagents: Sodium bicarbonate Solvents: Water; rt	
Experimental Protocols		

31-320-CAS-10279041	Steps: 1 Yield: 97%	Application of tridentate bis(oxazoline) ligands in catalytic asymmetric Nozaki-Hiyama allylation and crotylation: an example of high enantioselection with a non-symmetric bis(oxazoline) ligand By: McManus, Helen A.; et al Advanced Synthesis & Catalysis (2006), 348(4 + 5), 551-558.
1.1	Reagents: Manganese Catalysts: Chromium chloride (CrCl ₃) Solvents: Acetonitrile, Tetrahydrofuran; 10 min; 60 min, rt	
1.2	Reagents: Diisopropylethylamine Catalysts: 2-[(4 <i>S</i>)-4,5-Dihydro-4-(1-methylethyl)-2-oxazolyl]- <i>N</i> -[2-[(4 <i>S</i>)-4-(1,1-dimethylethyl)-4,5-dihydro-2-oxazolyl]phenyl] benzenamine; 1 h, rt	
1.3	1 h, rt	
1.4	Reagents: Chlorotrimethylsilane; 16 h, rt	
1.5	Reagents: Sodium bicarbonate Solvents: Water	
1.6	Reagents: Hydrochloric acid Solvents: Tetrahydrofuran, Water; 5 min	

Scheme 20 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-41431826

Steps: 1 Yield: 96%

1.1 **Reagents:** Hantzsch ester, Diisopropylethylamine, Zinc iodide
Catalysts: Dibromo[1,2-di(methoxy-κO)ethane]nickel, 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile, 3054393-65-3
Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C

Experimental Protocols

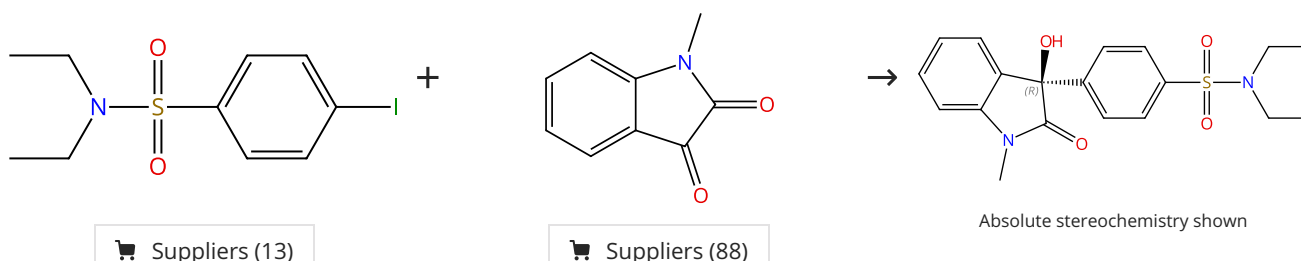
Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis

By: Gu, Pei; et al

Angewandte Chemie, International Edition (2024), 63(37), e202408195.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-44352330

Steps: 1 Yield: 96%

1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron
Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]
Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

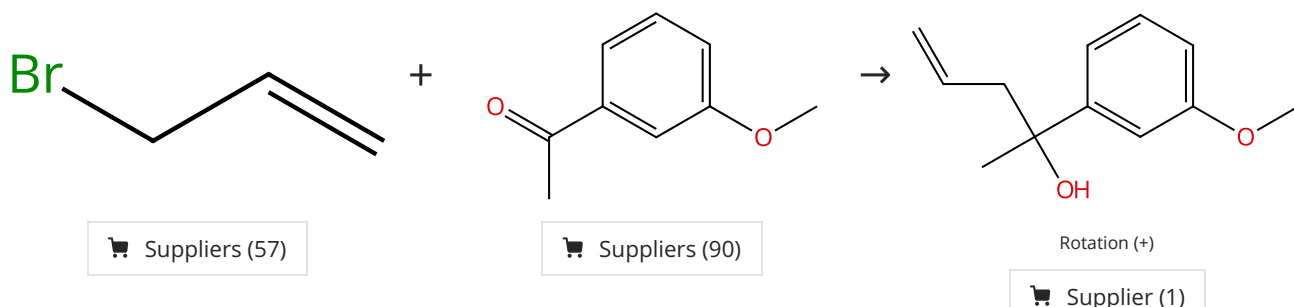
Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

Chinese Chemical Letters (2025), 36(5), 110243.

Scheme 22 (1 Reaction)

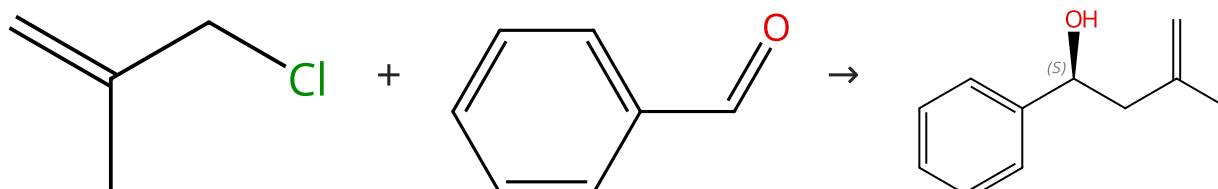
Steps: 1 Yield: 96%



31-320-CAS-3550298	Steps: 1 Yield: 96%	A Chiral Bipyridyl Alcohol for Catalytic Enantioselective Nozaki-Hiyama-Kishi Allylation of Aldehydes and Ketones By: Chen, Rui-Yu; et al Advanced Synthesis & Catalysis (2015), 357(5), 961-966.
1.1 Reagents: Manganese Catalysts: Chromium chloride (CrCl ₃) Solvents: Tetrahydrofuran; 1 h, rt		
1.2 Reagents: Triethylamine Catalysts: (5 <i>S</i> ,7 <i>S</i> ,8 <i>R</i>)-5,6,7,8-Tetrahydro-6,6-dimethyl- α,α -diphenyl-2-(2-pyridinyl)-5,7-methanoquinoline-8-methanol; 1 h, rt		
1.3 1 h, rt		
1.4 Reagents: Chlorotrimethylsilane; 36 h, rt		
1.5 Reagents: Sodium bicarbonate Solvents: Water		
1.6 Reagents: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; rt; 15 min, rt		
1.7 Reagents: Water		
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (58)

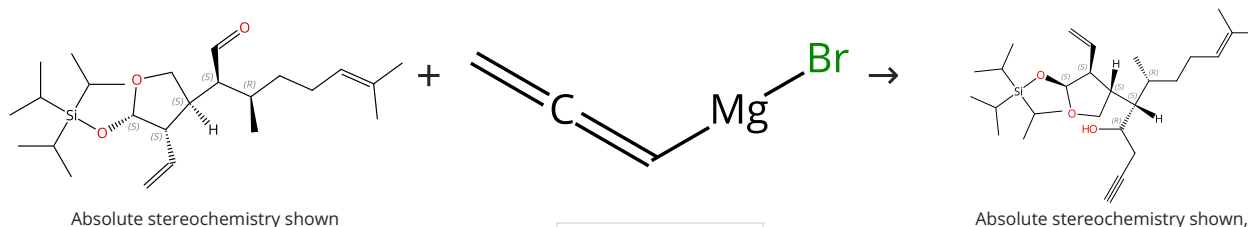
Suppliers (69)

Suppliers (2)

31-320-CAS-23455455	Steps: 1 Yield: 96%	Chromium(II) chloride By: Takai, Kazuhiko e-EROS Encyclopedia of Reagents for Organic Synthesis (2016), 1-19.
1.1 Reagents: Diisopropylethylamine, Manganese Catalysts: Chromium dichloride, 1,8-Bis[(4 <i>S</i>)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]-3,6-diphenyl-9 <i>H</i> -carbazole Solvents: Tetrahydrofuran; rt		
1.2 rt		
1.3 Reagents: Chlorotrimethylsilane		
1.4 Reagents: Tetrabutylammonium fluoride		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown

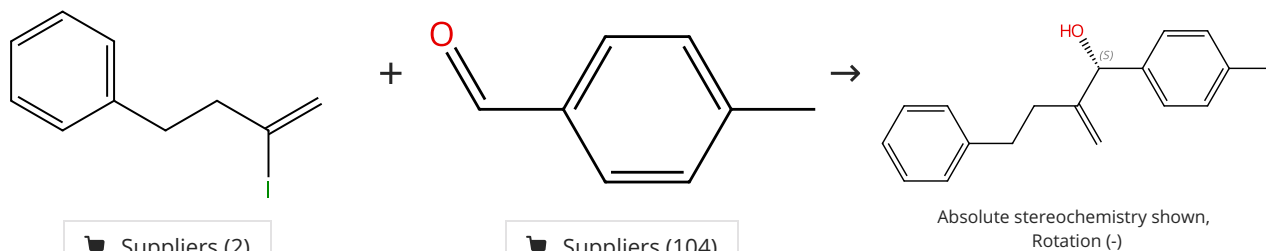
Suppliers (4)

Absolute stereochemistry shown, Rotation (+)

31-320-CAS-1895854	Steps: 1 Yield: 96%	Studies for the Synthesis of Xenicane Diterp enes. A Stereocntrrolled Total Synthesis of 4-Hydroxydictyolactone By: Williams, David R.; et al Journal of the American Chemical Society (2009), 131(25), 9038-9045.
1.1 Solvents: Diethyl ether; 30 min, -20 °C; 2 h, -20 °C		
1.2 Reagents: Ammonium chloride Solvents: Water; -20 °C		
Experimental Protocols		

Scheme 25 (2 Reactions)

Steps: 1 Yield: 96%



31-614-CAS-41431878

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Hantzsch ester, Diisopropylethylamine, Zinc iodide
Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile, 3054393-59-5
Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C

Experimental Protocols

Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis

By: Gu, Pei; et al

Angewandte Chemie, International Edition (2024), 63(37), e202408195.

31-614-CAS-41431827

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Hantzsch ester, Diisopropylethylamine, Zinc iodide
Catalysts: Dibromo[1,2-di(methoxy-κ*O*)ethane]nickel, 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile, 3054393-65-3
Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C

Experimental Protocols

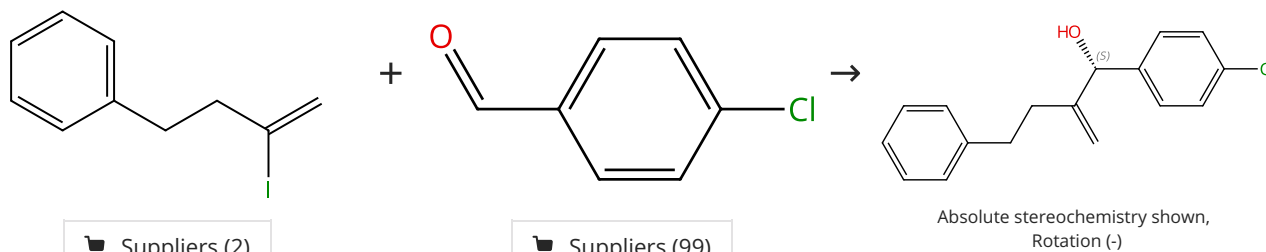
Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis

By: Gu, Pei; et al

Angewandte Chemie, International Edition (2024), 63(37), e202408195.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 95%



31-614-CAS-41431844

Steps: 1 Yield: 95%

- 1.1 **Reagents:** Hantzsch ester, Diisopropylethylamine, Zinc iodide
Catalysts: Dibromo[1,2-di(methoxy-κ*O*)ethane]nickel, 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile, 3054393-65-3
Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C

Experimental Protocols

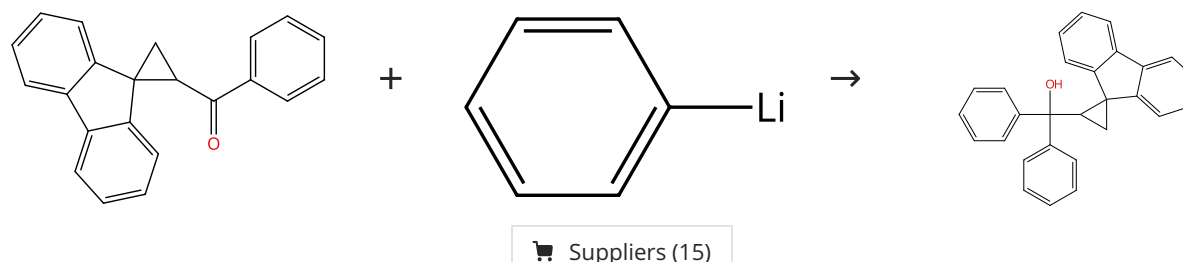
Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis

By: Gu, Pei; et al

Angewandte Chemie, International Edition (2024), 63(37), e202408195.

Scheme 27 (1 Reaction)

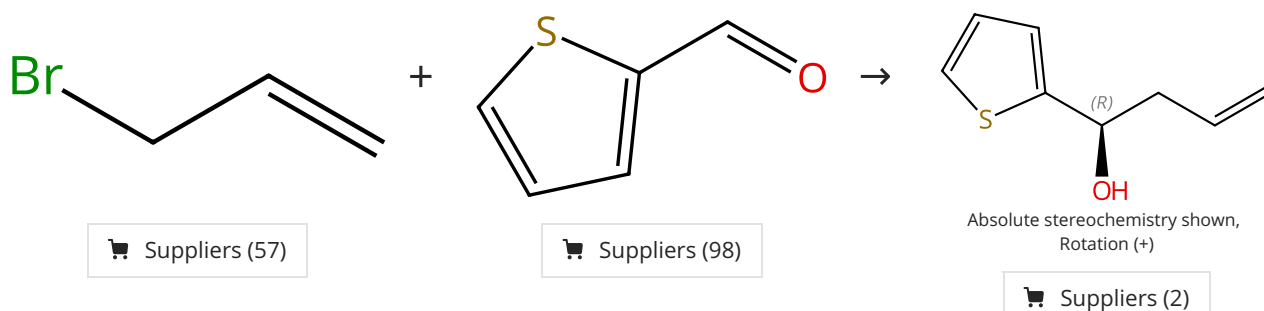
Steps: 1 Yield: 95%



31-320-CAS-22304511 Steps: 1 Yield: 95% 1.1 Solvents: Diethyl ether, Butyl ether; 5 min, - 78 °C; 1 h, - 78 °C; - 78 °C → 20 °C; 16 h, 20 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Evidence against Single-Electron Transfer in the Additions of Most Organomagnesium Reagents to Carbonyl Compounds By: Bartolo, Nicole D.; et al Journal of Organic Chemistry (2020), 85(12), 7848-7862.
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Scheme 28 (1 Reaction)

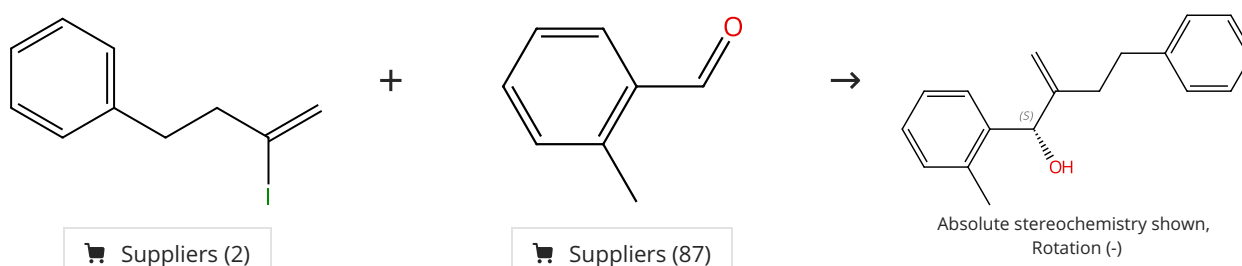
Steps: 1 Yield: 95%



31-320-CAS-10631915 Steps: 1 Yield: 95% 1.1 Reagents: Manganese Catalysts: Chromium chloride (CrCl₃) Solvents: Tetrahydrofuran; 1 h, rt 1.2 Reagents: Triethylamine Catalysts: (5<i>S</i>,7<i>S</i>,8<i>R</i>)-5,6,7,8-Tetrahydro-6,6-dimethyl-α,α-diphenyl-2-(2-pyridinyl)-5,7-methanoquinoline-8-methanol; 1 h, rt 1.3 1 h, rt 1.4 Reagents: Chlorotrimethylsilane; 36 h, rt 1.5 Reagents: Sodium bicarbonate Solvents: Water 1.6 Reagents: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; rt; 15 min, rt 1.7 Reagents: Water Experimental Protocols	A Chiral Bipyridyl Alcohol for Catalytic Enantioselective Nozaki-Hiyama-Kishi Allylation of Aldehydes and Ketones By: Chen, Rui-Yu; et al Advanced Synthesis & Catalysis (2015), 357(5), 961-966.
--	---

Scheme 29 (1 Reaction)

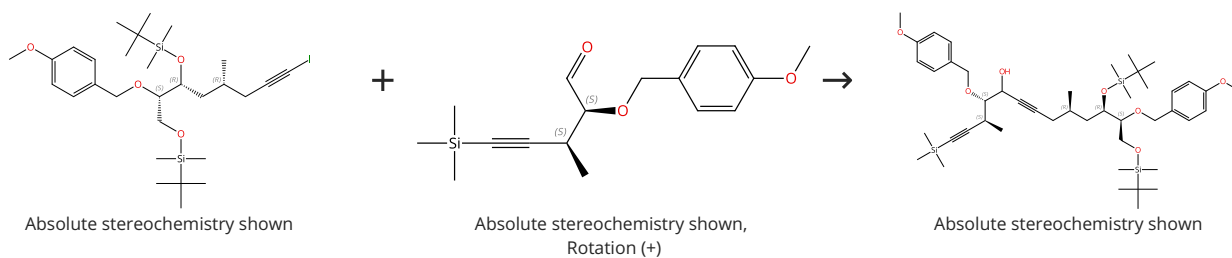
Steps: 1 Yield: 95%



31-614-CAS-41431829 Steps: 1 Yield: 95% 1.1 Reagents: Hantzsch ester, Diisopropylethylamine, Zinc iodide Catalysts: Dibromo[1,2-di(methoxy-κO)ethane]nickel, 2,4,5,6-Tetra(9<i>H</i>-carbazol-9-yl)isophthalonitrile, 3054393-65-3 Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C Experimental Protocols	Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis By: Gu, Pei; et al Angewandte Chemie, International Edition (2024), 63(37), e202408195.
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Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



31-320-CAS-8832259

Steps: 1 Yield: 95%

Synthesis of the C22-C40 Domain of the Azaspiracids

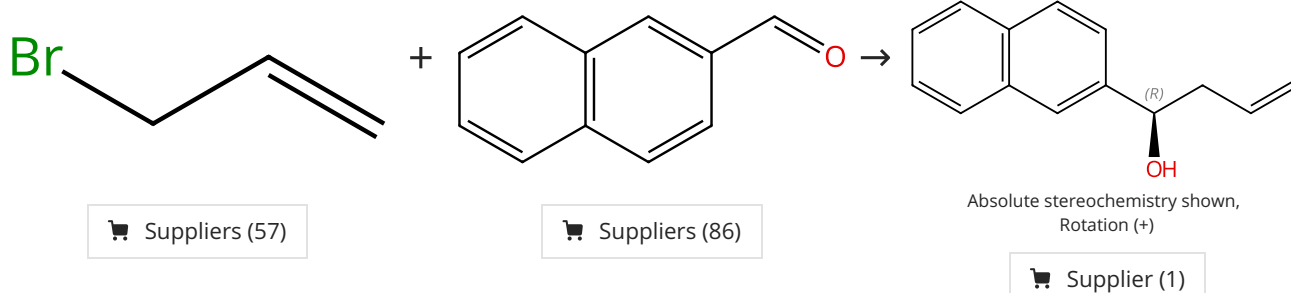
- 1.1 Reagents: Chromium dichloride
Catalysts: Nickel dichloride
Solvents: Tetrahydrofuran; 14 h, rt
- 1.2 Reagents: Sodium bicarbonate
Solvents: Diethyl ether, Water; 2 h, rt

By: Zhang, Zhigao; et al
Organic Letters (2016), 18(8), 1824-1827.

Experimental Protocols

Scheme 31 (1 Reaction)

Steps: 1 Yield: 95%



31-320-CAS-4589991

Steps: 1 Yield: 95%

A Chiral Bipyridyl Alcohol for Catalytic Enantioselective Nozaki-Hiyama-Kishi Allylation of Aldehydes and Ketones

- 1.1 Reagents: Manganese
Catalysts: Chromium chloride (CrCl₃)
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 Reagents: Triethylamine
Catalysts: (5*S*,7*S*,8*R*)-5,6,7,8-Tetrahydro-6,6-dimethyl- α,α -diphenyl-2-(2-pyridinyl)-5,7-methanoquinoline-8-methanol; 1 h, rt
- 1.3 1 h, rt
- 1.4 Reagents: Chlorotrimethylsilane; 36 h, rt
- 1.5 Reagents: Sodium bicarbonate
Solvents: Water
- 1.6 Reagents: Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; rt; 15 min, rt
- 1.7 Reagents: Water

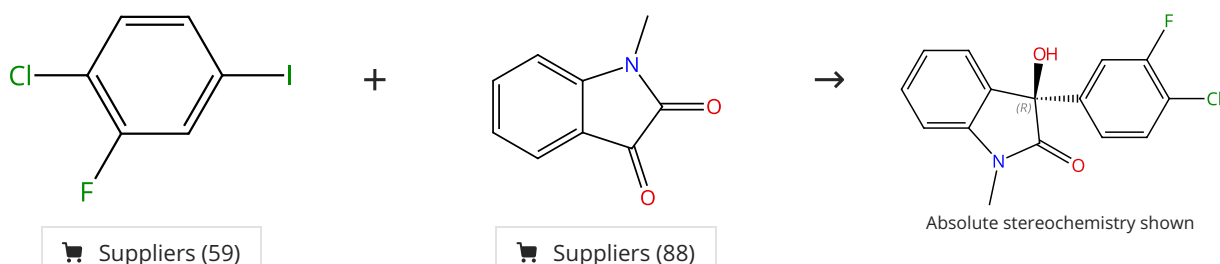
By: Chen, Rui-Yu; et al

Advanced Synthesis & Catalysis (2015), 357(5), 961-966.

Experimental Protocols

Scheme 32 (1 Reaction)

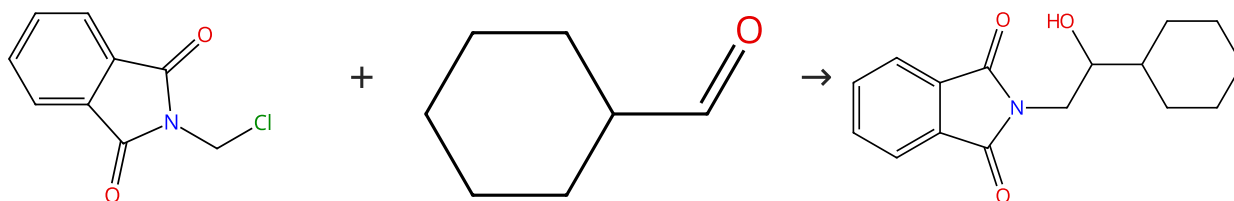
Steps: 1 Yield: 95%



31-614-CAS-44352321	Steps: 1 Yield: 95%	Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis
1.1 Reagents: Tripotassium phosphate, Bis(neopentyl glycolato) diboron Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κP)ferrocene] dichloro-, (7-4)-, (1 <i>R</i> ,1' <i>R</i>)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine] Solvents: Mesitylene; 24 h, 85 °C		By: Xiao, Jieshuai; et al Chinese Chemical Letters (2025), 36(5), 110243.
Experimental Protocols		

Scheme 33 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (67)

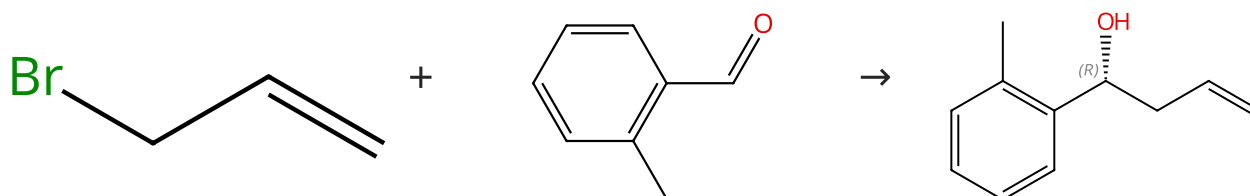
Suppliers (81)

Suppliers (5)

31-320-CAS-23456683	Steps: 1 Yield: 95%	Chromium(II) chloride
1.1 Reagents: Chromium dichloride, Lithium iodide Solvents: Tetrahydrofuran; 50 °C		By: Takai, Kazuhiko e-EROS Encyclopedia of Reagents for Organic Synthesis (2016), 1-19.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (57)

Suppliers (87)

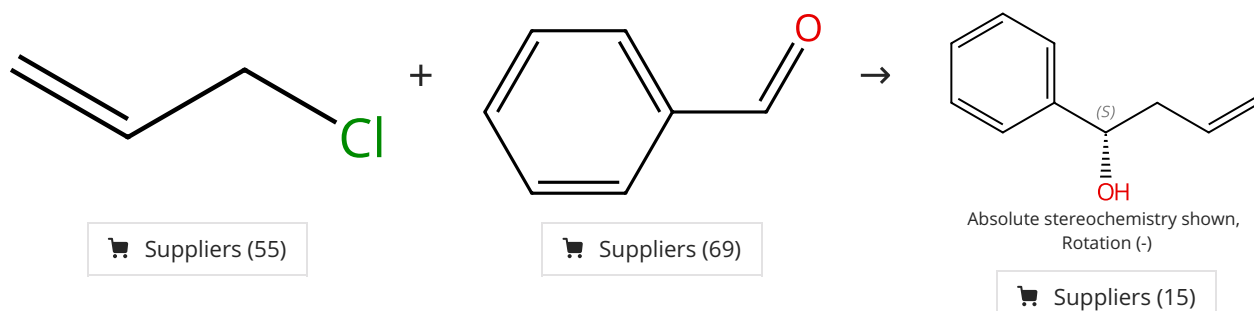
 Absolute stereochemistry shown,
Rotation (+)

Supplier (1)

31-320-CAS-9485492	Steps: 1 Yield: 95%	A pineno-salen type catalyst for the enantioselective Nozaki-Hiyama-Kishi reaction
1.1 Reagents: Manganese Solvents: Tetrahydrofuran; 1 h, rt		By: Dhondge, Attrimuni P.; et al
1.2 Catalysts: Triethylamine, 2,4-Bis(1,1-dimethylethyl)-6-[(<i>E</i>)-[[[(5 <i>S</i> ,7 <i>R</i> ,8 <i>S</i>)-5,6,7,8-tetrahydro-6,6-dimethyl-2-phenyl-5,7-methanoquinolin-8-yl]imino]methyl]phenol; 1 h, rt		Tetrahedron: Asymmetry (2012), 23(10), 723-733.
1.3 1 h, rt		
1.4 Reagents: Chlorotrimethylsilane; 20 h, rt		
1.5 Reagents: Sodium bicarbonate Solvents: Water; rt		
1.6 Reagents: Hydrochloric acid Solvents: Tetrahydrofuran; rt; 30 min, rt		
1.7 Reagents: Sodium bicarbonate Solvents: Water; rt		
Experimental Protocols		

Scheme 35 (1 Reaction)

Steps: 1 Yield: 95%



31-320-CAS-23454938

Steps: 1 Yield: 95%

Chromium(II) chloride

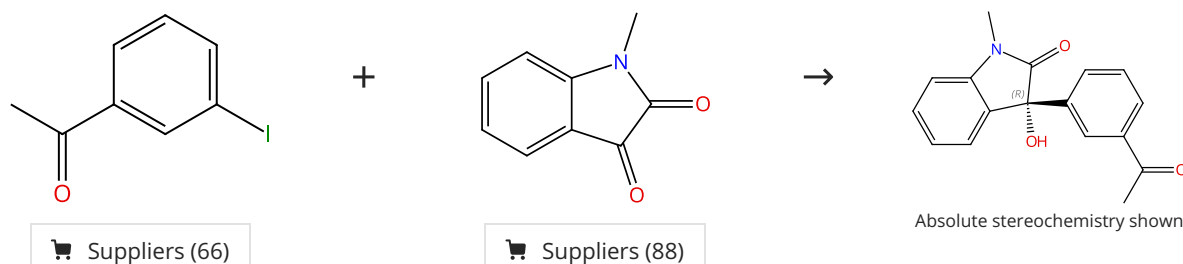
By: Takai, Kazuhiko

e-EROS Encyclopedia of Reagents for Organic Synthesis (2016), 1-19.

- 1.1 **Reagents:** Diisopropylethylamine, Manganese
Catalysts: Chromium dichloride, 1,8-Bis[(4*S*)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]-3,6-diphenyl-9*H*-carbazole
Solvents: Tetrahydrofuran; rt
- 1.2 rt
- 1.3 **Reagents:** Chlorotrimethylsilane
- 1.4 **Reagents:** Tetrabutylammonium fluoride

Scheme 36 (1 Reaction)

Steps: 1 Yield: 94%



31-614-CAS-44352333

Steps: 1 Yield: 94%

Asymmetric Nozaki-Hiyama-Kishi (NHK)-type reaction of isatins with aromatic iodides by cobalt catalysis

By: Xiao, Jieshuai; et al

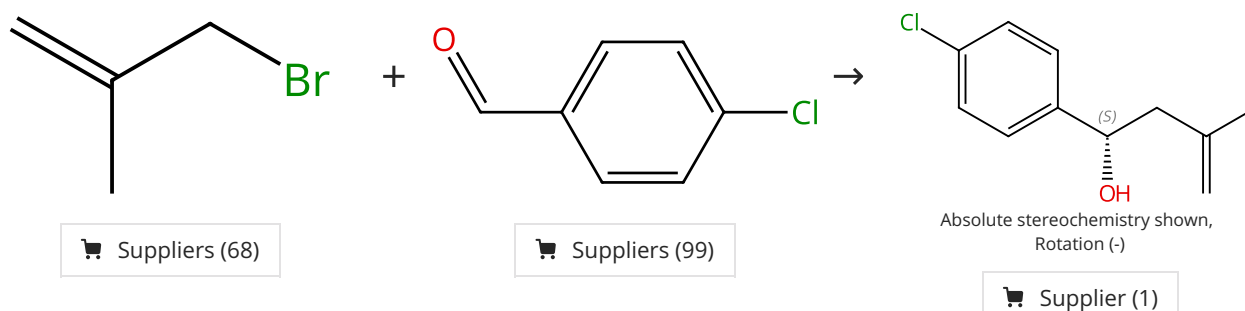
Chinese Chemical Letters (2025), 36(5), 110243.

- 1.1 **Reagents:** Tripotassium phosphate, Bis(neopentyl glycolato) diboron
Catalysts: Cobalt, [1,1'-bis(diphenylphosphino-κ*P*)ferrocene] dichloro-, (*T*-4)-, (1*R*,1'*R*)-1,1'-(1,2-Phenylene)bis[1-(1,1-dimethylethyl)-1-methylphosphine]
Solvents: Mesitylene; 24 h, 85 °C

Experimental Protocols

Scheme 37 (1 Reaction)

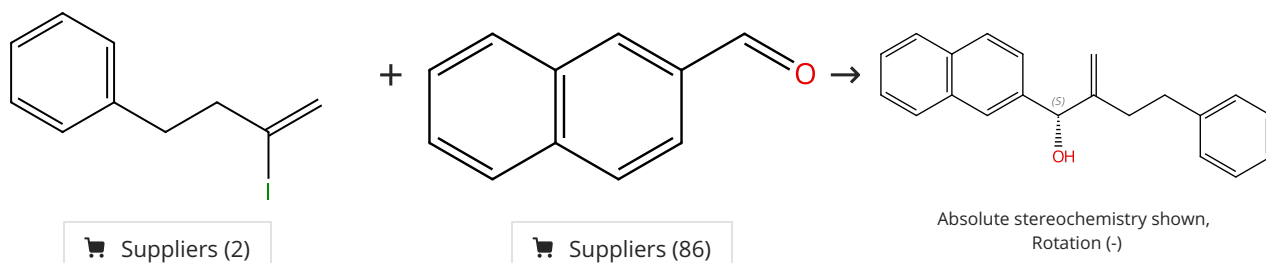
Steps: 1 Yield: 94%



31-320-CAS-8242596 Steps: 1 Yield: 94% 1.1 Reagents: Manganese Catalysts: Chromium dichloride, (1<i>Z</i>,3<i>Z</i>)-1,3-Bis[[<i>(4S)</i>-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]methylene]-2,3-dihydro-5,6-dimethyl-1<i>H</i>-isoindole Solvents: Tetrahydrofuran; 1 h, rt 1.2 Catalysts: Diisopropylethylamine; 30 min, rt 1.3 1 h, rt 1.4 Reagents: Chlorotrimethylsilane; 12 h, rt 1.5 Reagents: Sodium bicarbonate Solvents: Water; rt 1.6 Reagents: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; 30 min, rt 1.7 Reagents: Ammonium chloride Solvents: Water; rt Experimental Protocols	The Synthesis of a New Class of Chiral Pincer Ligands and Their Applications in Enantioselective Catalytic Fluorinations and the Nozaki-Hiyama-Kishi Reaction By: Deng, Qing-Hai; et al Chemistry - A European Journal (2011), 17(52), 4922-14928, S14922/1-S14922/133.
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Scheme 38 (1 Reaction)

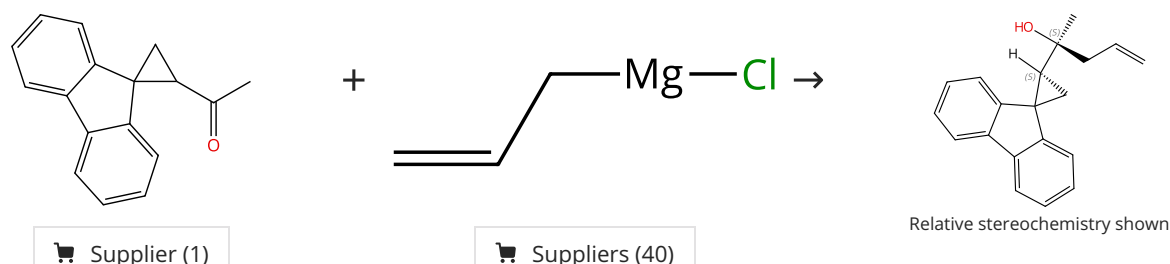
Steps: 1 Yield: 94%



31-614-CAS-41431850 Steps: 1 Yield: 94% 1.1 Reagents: Hantzsch ester, Diisopropylethylamine, Zinc iodide Catalysts: Dibromo[1,2-di(methoxy-κO)ethane]nickel, 2,4,5,6-Tetra(9<i>H</i>-carbazol-9-yl)isophthalonitrile, 3054393-65-3 Solvents: Tetrahydrofuran, Dimethylacetamide; 24 h, 30 °C Experimental Protocols	Chromium- and Metal-Reductant-Free Asymmetric Nozaki-Hiyama-Kishi (NHK) Reaction Enabled by Metallaphotoredox Catalysis By: Gu, Pei; et al Angewandte Chemie, International Edition (2024), 63(37), e202408195.
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Scheme 39 (1 Reaction)

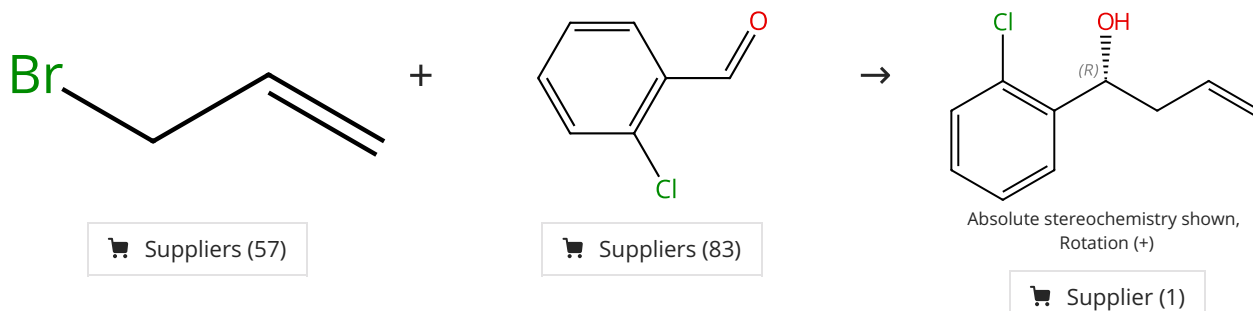
Steps: 1 Yield: 94%



31-320-CAS-22304497 Steps: 1 Yield: 94% 1.1 Solvents: Tetrahydrofuran; 5 min, -78 °C; 30 min, -78 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	Evidence against Single-Electron Transfer in the Additions of Most Organomagnesium Reagents to Carbonyl Compounds By: Bartolo, Nicole D.; et al Journal of Organic Chemistry (2020), 85(12), 7848-7862.
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Scheme 40 (1 Reaction)

Steps: 1 Yield: 94%



31-320-CAS-5402908

Steps: 1 Yield: 94%

- 1.1 **Reagents:** Manganese
Catalysts: Chromium chloride (CrCl_3)
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 **Reagents:** Triethylamine
Catalysts: (5*S*,7*S*,8*R*)-5,6,7,8-Tetrahydro-6,6-dimethyl- α , α -diphenyl-2-(2-pyridinyl)-5,7-methanoquinoline-8-methanol; 1 h, rt
- 1.3 1 h, rt
- 1.4 **Reagents:** Chlorotrimethylsilane; 36 h, rt
- 1.5 **Reagents:** Sodium bicarbonate
Solvents: Water
- 1.6 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; rt; 15 min, rt
- 1.7 **Reagents:** Water

Experimental Protocols

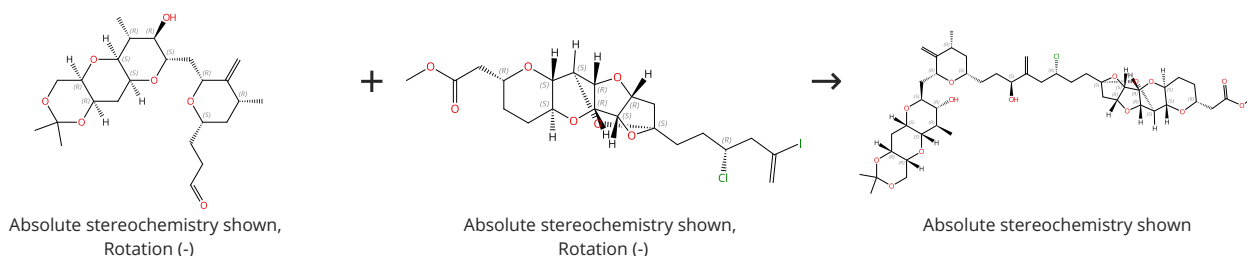
A Chiral Bipyridyl Alcohol for Catalytic Enantioselective Nozaki-Hiyama-Kishi Allylation of Aldehydes and Ketones

By: Chen, Rui-Yu; et al

Advanced Synthesis & Catalysis (2015), 357(5), 961-966.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 94%



31-614-CAS-43646467

Steps: 1 Yield: 94%

- 1.1 **Reagents:** Zirconocene, dichloride, Manganese, Lithium chloride, Proton sponge
Catalysts: Chromium dichloride, 3,5-Dichloro-*N*-[2-[(4*R*)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]-6-[[[(3*R*,5*S*)-3,5-dimethylcyclohexyl]oxy]phenyl]benzenesulfonamide, Dichloro(2,9-diethyl-1,10-phenanthroline- κ^1 , κ^1)nickel
Solvents: Acetonitrile; rt

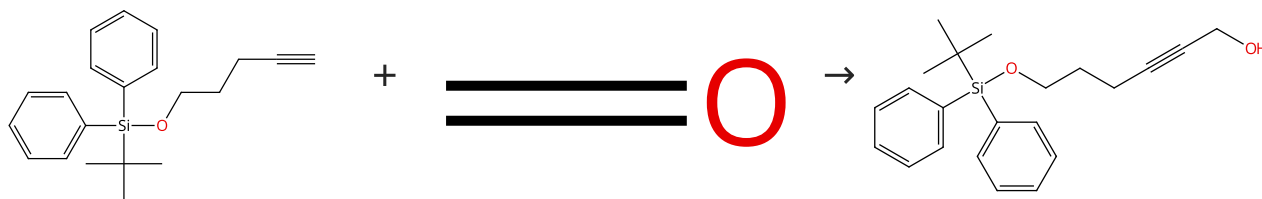
What Does It Take to Develop Structurally Complex Molecules by Total Synthesis? Rapid Process Development and GMP Manufacturing of E7130 Drug Substance for First-in-Human Clinical Study

By: Sasaki, Takeo; et al

Organic Process Research & Development (2024), 28(6), 2077-2089.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (4)

Suppliers (188)

Suppliers (2)

31-320-CAS-7740633

Steps: 1 Yield: 94%

Total Synthesis of the Protein Phosphatase 2A Inhibitor Lactodehydrothysiferol

By: Clausen, Dane J.; et al

Angewandte Chemie, International Edition (2011), 50(22), 5178-5181.

1.1 Reagents: Butyllithium

Solvents: Tetrahydrofuran, Hexane; -78 °C; 1.5 h, -78 °C

1.2 -78 °C → rt; overnight, rt

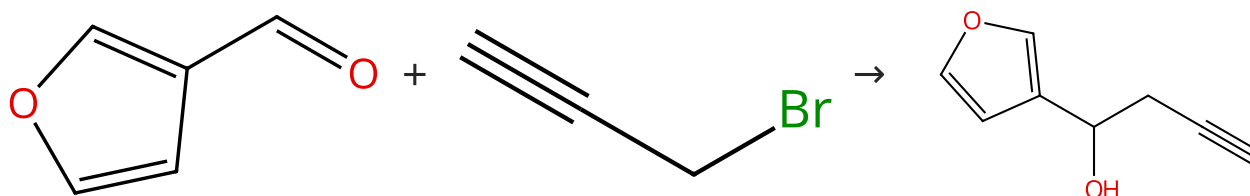
1.3 Reagents: Ammonium chloride

Solvents: Water

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (84)

Suppliers (73)

Suppliers (6)

31-320-CAS-18334476

Steps: 1 Yield: 94%

Scalable Regioselective and Stereoselective Synthesis of Functionalized (E)-4-Iodobut-3-en-1-ols: Gram-Scale Total Synthesis of Fungal Decanolides and Derivatives

By: Sherwood, Alexander M.; et al

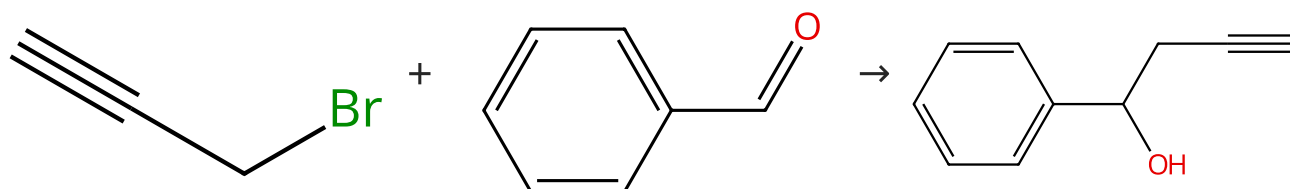
Journal of Organic Chemistry (2018), 83(2), 980-992.

1.1 Reagents: Zinc

Solvents: Diethyl ether, Dimethylformamide, Toluene; 10 min, rt; 12 h, rt

Scheme 44 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (73)

Suppliers (69)

Suppliers (53)

31-320-CAS-18333828

Steps: 1 Yield: 94%

Scalable Regioselective and Stereoselective Synthesis of Functionalized (E)-4-Iodobut-3-en-1-ols: Gram-Scale Total Synthesis of Fungal Decanolides and Derivatives

By: Sherwood, Alexander M.; et al

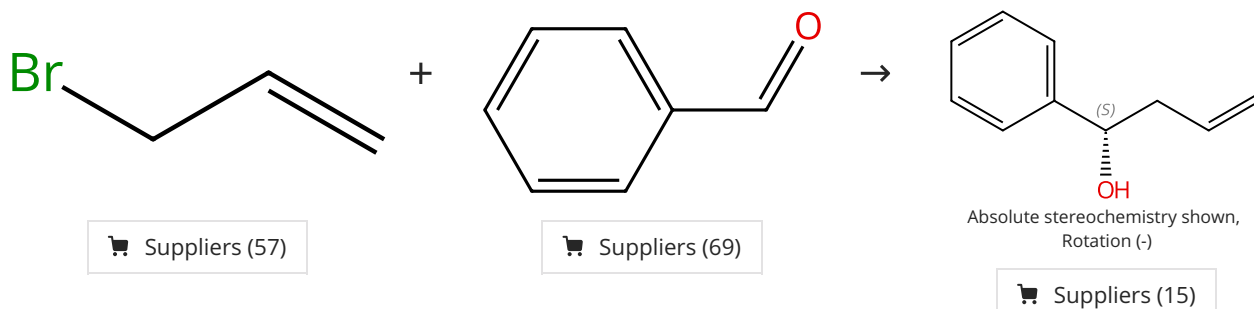
Journal of Organic Chemistry (2018), 83(2), 980-992.

1.1 Reagents: Zinc

Solvents: Diethyl ether, Dimethylformamide, Toluene; 10 min, rt; 12 h, rt

Scheme 45 (3 Reactions)

Steps: 1 Yield: 93%



31-320-CAS-23454929

Steps: 1 Yield: 93%

- 1.1 **Reagents:** Diisopropylethylamine, Manganese
Catalysts: Chromium dichloride, 1,8-Bis[(4*S*)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl]-3,6-diphenyl-9*H*-carbazole
Solvents: Tetrahydrofuran; rt
- 1.2 rt
- 1.3 **Reagents:** Chlorotrimethylsilane
- 1.4 **Reagents:** Tetrabutylammonium fluoride

Chromium(II) chloride

By: Takai, Kazuhiko

e-EROS Encyclopedia of Reagents for Organic Synthesis (2016), 1-19.

31-320-CAS-10710380

Steps: 1 Yield: 93%

- 1.1 **Reagents:** Manganese
Catalysts: Chromium dichloride, (1*Z*,3*Z*)-1,3-Bis[[[(4*S*)-4,5-dihydro-4-phenyl-2-oxazolyl]methylene]-2,3-dihydro-5,6-diphenyl-1*H*-isoindole
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 **Catalysts:** Diisopropylethylamine; 30 min, rt
- 1.3 1 h, rt
- 1.4 **Reagents:** Chlorotrimethylsilane; 12 h, rt
- 1.5 **Reagents:** Sodium bicarbonate
Solvents: Water; rt
- 1.6 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; 30 min, rt
- 1.7 **Reagents:** Ammonium chloride
Solvents: Water; rt

The Synthesis of a New Class of Chiral Pincer Ligands and Their Applications in Enantioselective Catalytic Fluorinations and the Nozaki-Hiyama-Kishi Reaction

By: Deng, Qing-Hai; et al

Chemistry - A European Journal (2011), 17(52), 4922-14928, S14922/1-S14922/133.

Experimental Protocols

31-320-CAS-6443087

Steps: 1 Yield: 93%

- 1.1 **Reagents:** Manganese
Catalysts: Chromium dichloride, (1*Z*,3*Z*)-1,3-Bis[[[(4*S*)-4-(1,1-dimethylethyl)-4,5-dihydro-2-oxazolyl]methylene]-2,3-dihydro-5,6-dimethyl-1*H*-isoindole
Solvents: Tetrahydrofuran; 1 h, rt
- 1.2 **Catalysts:** Diisopropylethylamine; 30 min, rt
- 1.3 1 h, rt
- 1.4 **Reagents:** Chlorotrimethylsilane; 10 h, rt
- 1.5 **Reagents:** Sodium bicarbonate
Solvents: Water; rt
- 1.6 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; 30 min, rt
- 1.7 **Reagents:** Ammonium chloride
Solvents: Water; rt

The Synthesis of a New Class of Chiral Pincer Ligands and Their Applications in Enantioselective Catalytic Fluorinations and the Nozaki-Hiyama-Kishi Reaction

By: Deng, Qing-Hai; et al

Chemistry - A European Journal (2011), 17(52), 4922-14928, S14922/1-S14922/133.

Experimental Protocols

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